

Data Intake Processing and Verification Report

Background Information

- **Original Dataset Name:** Pilot top-down CO₂ budget constrained by the v10 OCO-2 MIP Version 1
- **GHG Center Dataset Title:** OCO-2 Modeling Intercomparison Project (MIP) CO₂ Flux
 - Title prior to September 2025: OCO-2 MIP Top-down CO₂ Budget
- **Dataset Provider:** CEOS, NASA/JPL
- **Date Obtained:** August 2023
- **Location Obtained From:** <https://doi.org/10.48588/npf6-sw92>
- **Data Location in GHG Center:** oco2-mip-co2budget-yeargrid-v1
- **Data POC(s):** Dr. Kevin Bowman, Dr. Matthew Thill, Dr. Brendan Byrne
- **Dataset File Type(s):** NetCDF
- **Projection (if different from WGS84):** NA

Data Transfer Confirmation

An SHA-256 checksum is used to detect high-level errors within data transmissions. Results from individual checksum file comparisons of pre-transfer and post-transfer shows all files were transferred successfully and no individual files had any transfer issues.

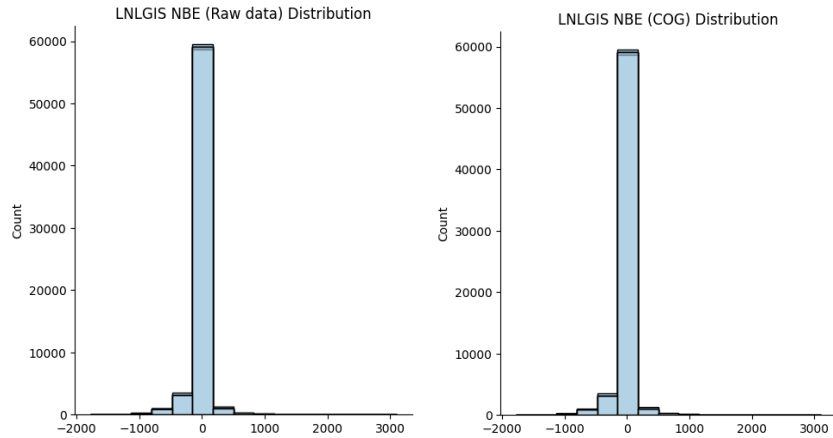
Data Intake Process

- https://us-ghg-center.github.io/ghgc-docs/data_workflow/oco2-mip-co2budget-yeargrid-v1_Data_Flow.html

Overall Dataset Statistics

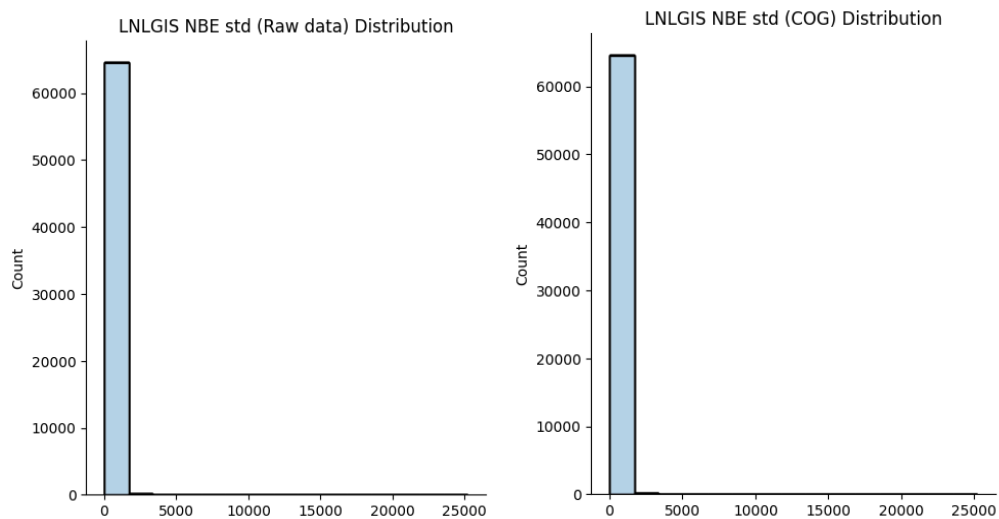
NetCDF (original data) and COGs (transformed cloud optimized GeoTIFF) files were read to generate the below comparison statistics.

- Statistics for LNLGIS Net Biosphere Exchange Variable:



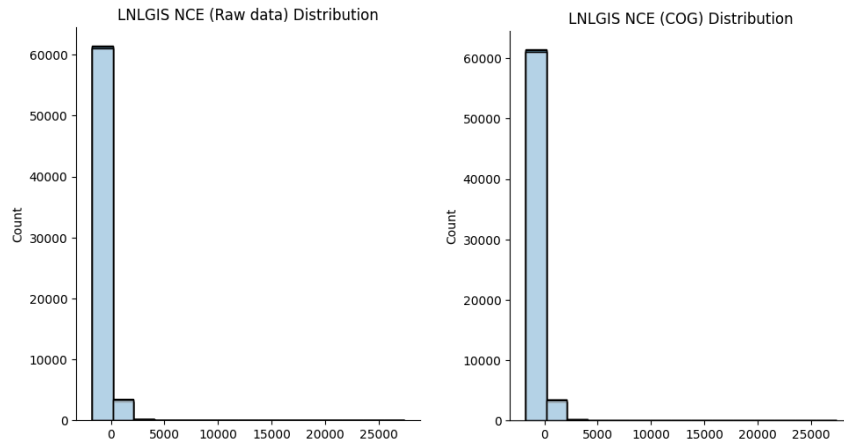
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-1781.954	3111.111	-18.466	138.722
Transformed Data	-1781.954	3111.111	-18.466	138.722

- Statistics for Uncertainty - LNLGIS Net Biosphere Exchange:



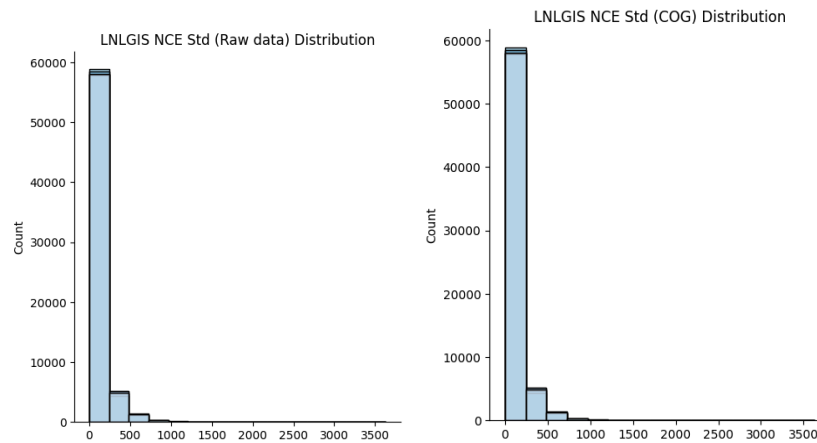
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	25217.181	76.933	272.433
Transformed Data	0.0	25217.181	76.933	272.433

- Statistics for LNLGIS Net Carbon Exchange variable (LNLGIS_NCE):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-1770.917	27399.8608	38.4327	438.115
Transformed Data	-1770.917	27399.8608	38.4327	438.115

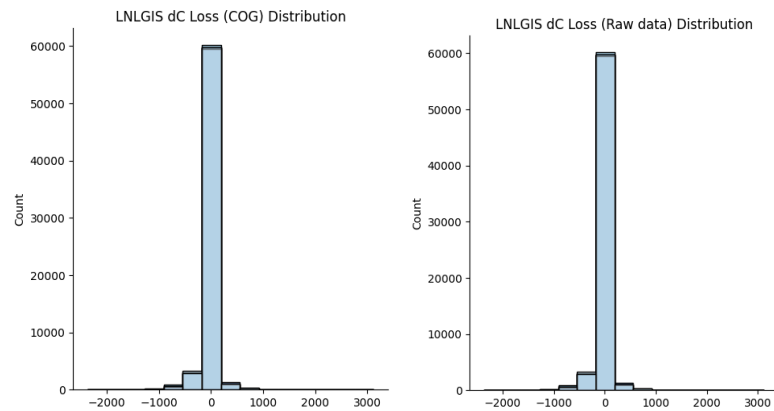
- Statistics for Uncertainty LNLGIS Net Carbon Exchange variable (LNLGIS_NCE_std):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	3632.335	59.0749	143.437
Transformed	0.0	3632.335	59.0749	143.437

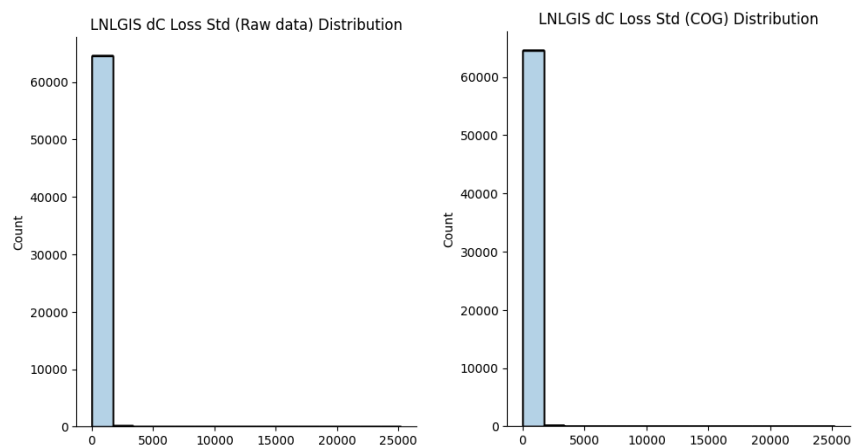
Data				
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- Statistics for LNLGIS Net Land Carbon Stock Loss variable (LNLGIS_dC_loss) :



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-2372.259	3122.9865	-16.1898	138.824
Transformed Data	-2372.259	3122.9865	-16.1898	138.824

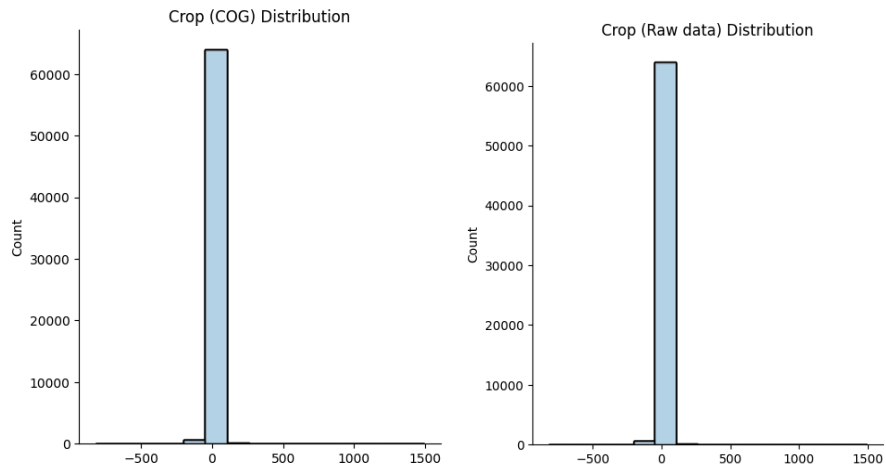
- Statistics for Uncertainty LNLGIS Net Land Carbon Stock Loss variable (LNLGIS_dC_loss_std):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	25217.188	77.09118	272.566

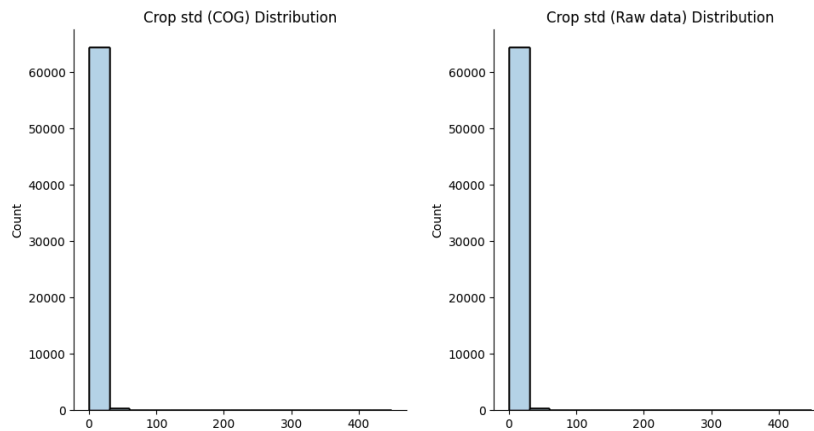
Transformed Data	0.0	25217.188	77.09118	272.566
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- Statistics for Lateral Crop CO2 variable (crop):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-821.231	1495.350	-0.0872	20.139
Transformed Data	-821.231	1495.350	-0.0872	20.139

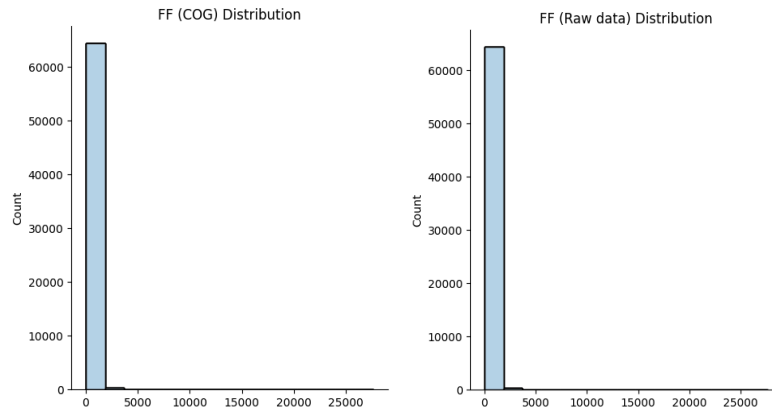
- Statistics for Uncertainty Later Crop CO2 variable (crop_std):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation

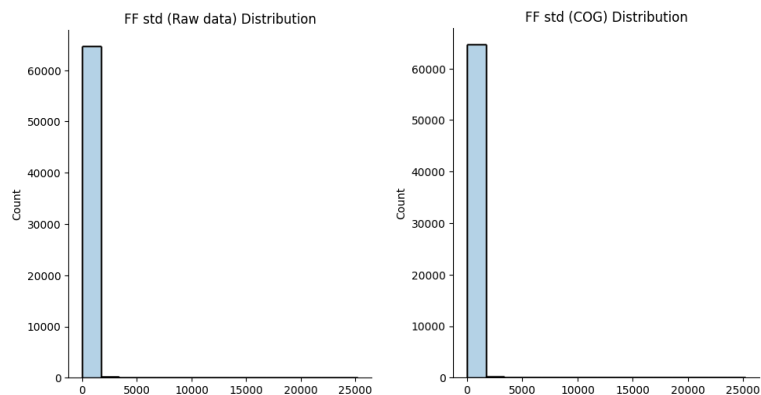
Original Data	0.0	448.605	0.950	5.966
Transformed Data	0.0	448.605	0.9508	5.966

- Statistics for Fossil Fuel and Cement CO2 Emissions variable (FF):



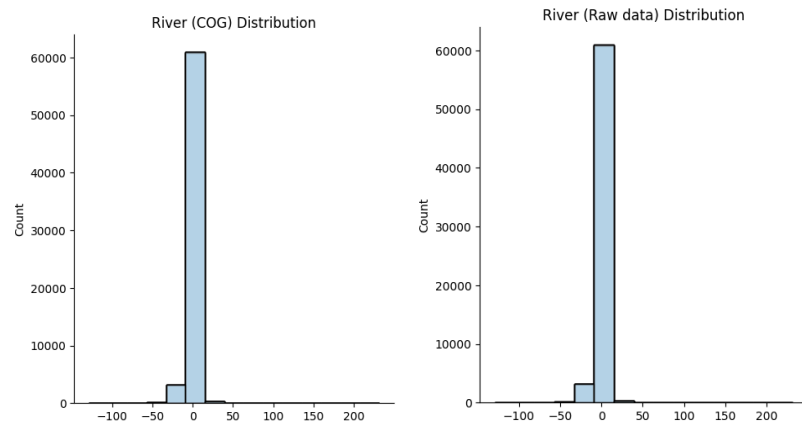
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	27670.273	56.898	446.918
Transformed Data	0.0	27670.273	56.898	446.918

- Statistics for Uncertainty Fossil Fuel and Cement CO2 Emissions variable (FF_std):



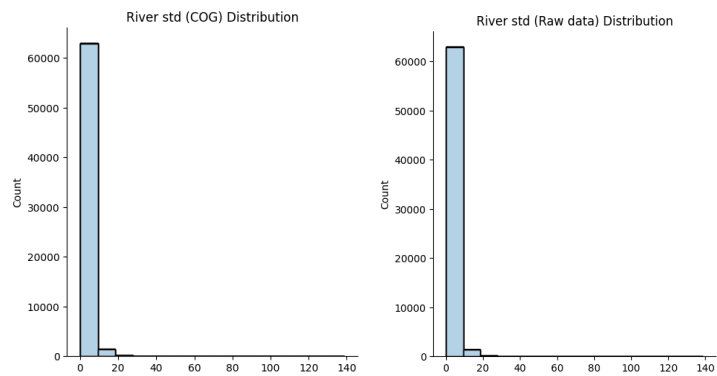
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	25215.769	27.169	235.236
Transformed Data	0.0	25215.769	27.169	235.236

- Statistics for Lateral River CO2 variable (River):



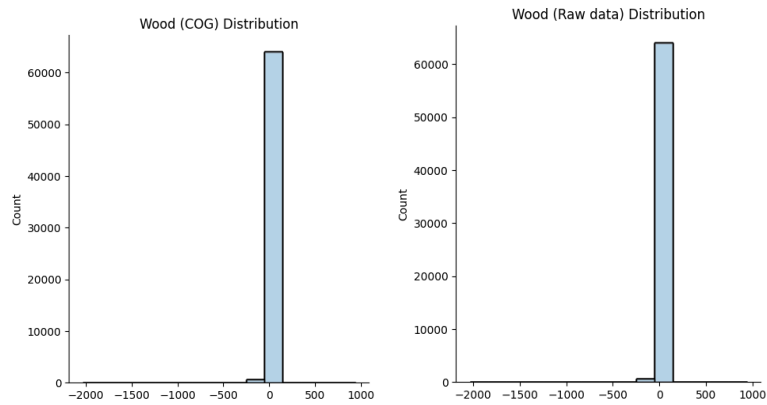
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-128.993	231.391	-0.9315	6.772
Transformed Data	-128.993	231.391	-0.9315	6.772

- Statistics for Uncertainty Lateral River CO2 variable (River_std):



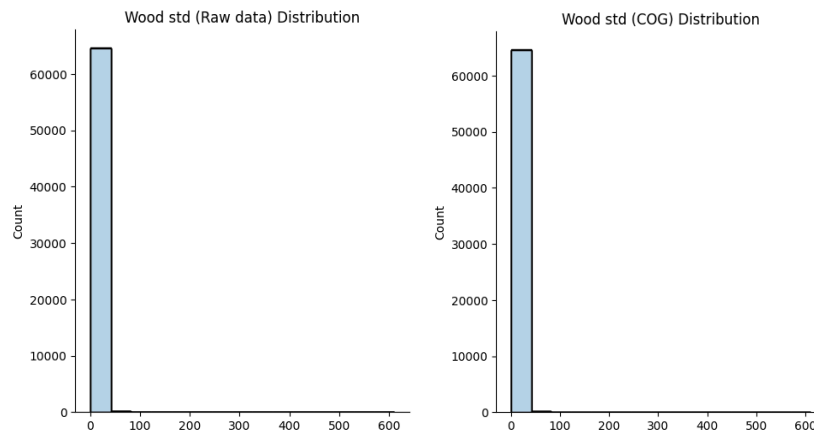
	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	138.834	1.1315	3.942
Transformed Data	0.0	138.834	1.1315	3.942

- Statistics for Lateral Wood CO2 Flux variable (Wood):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	-2036.632	937.564	-1.2575	21.9895
Transformed Data	-2036.632	937.564	-1.2575	21.9895

- Statistics for Uncertainty Lateral Wood CO2 Flux variable (Wood_std):



	Minimum (gCO ₂ /m ² /yr)	Maximum (gCO ₂ /m ² /yr)	Mean (gCO ₂ /m ² /yr)	Standard Deviation
Original Data	0.0	610.9896	0.9476	6.5393
Transformed Data	0.0	610.9896	0.9476	6.5393

- Link to transformation record in [Jupyter Notebook](#)
- All values are in expected range

Summary

- We are confident that the transformation and display of data in the GHG Center is correct
- There were no problems identified in the data
- [Link](#) to the transformation code
- [Link](#) to the dataset Tutorial
- [Link](#) to the US GHG Center data catalog page

Report Completed on: 11/3/2023; updated on 8/19/2025 to reflect updated dataset title
MSFC POC for questions: [Jeanné le Roux](#), [Siddharth Chaudhary](#)